

DATA SCRAPING AND EVALUATING NLP MODEL PERFORMANCE

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What is Data Scraping

Data scraping is the process of **automatically extracting structured or unstructured text data from web sources**. In NLP, scraping is essential because **high-quality text data is required for training and evaluating machine learning models**.

Why is Data Scraping Important for NLP?

- **Access to Large Datasets:** NLP models require large-scale text data to improve performance.
- **Automation:** Scraping eliminates manual data collection.
- **Custom Data Collection:** Allows researchers to **collect domain-specific text data** (e.g., product reviews, news articles, social media posts).
- **Real-time Insights:** Helps in applications like **sentiment analysis, market research, and fake news detection**.



Methods of Data Scraping

A. Web Scraping (Extracting HTML Data)

- . Extracts text data directly from websites.
- . Tools: **BeautifulSoup (Python), Scrapy, Selenium.**
- . Example: Scraping news headlines from **CNN or BBC** for sentiment analysis.

Challenges:

- ✓ Websites may block scrapers (rate limits, bot detection).
- ✓ Unstructured HTML data requires **cleaning and parsing.**



Methods of Data Scraping

B. API-Based Data Scraping

- . Uses official **APIs** to collect structured text data.
- . Example: **Twitter API** for scraping tweets.
- . API Tools: **Tweepy (Python), NewsAPI, Reddit API.**

Advantages:

- ✓ More structured and reliable than web scraping.
- ✓ Avoids ethical and legal issues.



Methods of Data Scrapping

C. Public Datasets & Open Data Repositories

- Platforms like **Kaggle**, **Google Dataset Search**, and **Hugging Face Datasets** offer pre-collected text data.
- Example:
 - **IMDB Sentiment Dataset** (for movie review sentiment analysis).
 - **Amazon Review Dataset** (for customer sentiment analysis).



Legal And Ethical Considerations In Data Scraping

- **Follow Website Policies:** Check robots.txt to see if scraping is allowed.
- **Avoid Private Data Collection:** Do not scrape sensitive user information.
- **Rate Limits & Requests:** Do not overload servers with too many requests.

Use Public APIs if Available: Prefer official API endpoints over direct web scraping.



Implementing Data Scrapping For NLP

Case Study: Scrapping Twitter Data for Sentiment Analysis

Objective

Scrape **tweets about a trending topic** (e.g., "Artificial Intelligence") and analyze public sentiment.

Step 1: Collect Data (Using Twitter API)

We extract the following **text attributes**:

- ✓ Tweet text
- ✓ Username (for reference)
- ✓ Timestamp (date & time)



Implementing Data Scrapping For NLP

Example of a scraped tweet dataset :

TWEET ID	USERNAME	TWEET TEXT	TIMESTAMP
001	@techguru	"AI is changing everything! #innovation"	2024-02-10 12:30
002	@datascientist	"Machine learning models are getting better at NLP!"	2024-02-10 12:45
003	@randomuser	"Not sure if AI will take my job or help me..."	2024-02-10 13:00

Step 2: Preprocessing Text Data

After scraping, the data needs to be cleaned before NLP analysis.

Key text preprocessing steps:

- 1.Convert text to lowercase → “AI is powerful” → “ai is powerful”
- 2.Remove stopwords → “AI is very powerful” → “AI powerful”
- 3.Tokenization → Splitting text into words
- 4.Stemming & Lemmatization → “learning” → “learn”
- 5.Removing special characters and punctuation

✓ Final Processed Data:

TWEET TEXT	CLEANED TEXT
"AI is changing everything! #innovation"	"ai changing everything"
"Machine learning models are getting better at NLP!"	"machine learn model better NLP"

EVALUATING NLP MODEL PERFORMANCE

A. What is Model Evaluation?

After training an NLP model, we must assess how well it performs. **Evaluation metrics** help measure the model's effectiveness in predicting the correct outputs.



B. Key Evaluation Metrics in NLP

METRIC	DEFINITION	USE CASE
ACCURACY	% of correct predictions	Sentiment Analysis
PRECISION	% of true positives among predicted positives	Fake News Detection
RECALL	% of actual positives correctly identified	Spam Detection
F1-SCORE	Balance between Precision & Recall	Text Classification
BLEU SCORE	Measures similarity to a reference text	Machine Translation
ROUGE SCORE	Compares generated text with actual text	Text Summarization
PERPLECITY	Measures fluency of a language model	Chatbot Performance

C. CASE STUDY : SENTIMENT ANALYSIS ON TWITTER DATA

DATASET :

- **Source:** Scraped Twitter data about "AI in Education".
- **Classes:** Positive, Negative, Neutral.

MACHINE LEARNING MODELS USED :

MODEL	ALGORITHM USED
Baseline Model	Naïve Bayes (Simple)
Advanced Model	BERT (Deep Learning)

EVALUATION RESULT

Model	Accuracy	Precision	Recall	F1-Score
Naïve Bayes	79%	76%	81%	78%
BERT	91%	90%	92%	91%

D. What is Model Evaluation?

- . **Naïve Bayes** is fast but struggles with sarcasm and negations.
- . **BERT** handles **context better**, leading to **higher accuracy**.
- . For **resource-limited applications**, Naïve Bayes is a **good choice**.
- . For **high-precision NLP tasks**, deep learning models like **BERT** are recommended.



STUDENT ASSIGNMENT : DATA SCRAPING & MODEL EVALUATION

Objective

Students will **scrape real-world text data**, clean it, and evaluate an NLP model's performance.

Task Instructions

1.Choose a Data Source

- Twitter (via API)
- News websites (via web scraping)
- Research papers (Google Scholar abstracts)

2.Scrape & Preprocess Data

- Collect at least **1000 text samples**.
- Clean the text (remove stopwords, punctuation, lowercase).



STUDENT ASSIGNMENT : DATA SCRAPING & MODEL EVALUATION

3. Train a Machine Learning Model

- Train a Naïve Bayes or SVM model for classification.

4. Evaluate Model Performance

- Calculate accuracy, precision, recall, and F1-score.

5. Write a Report

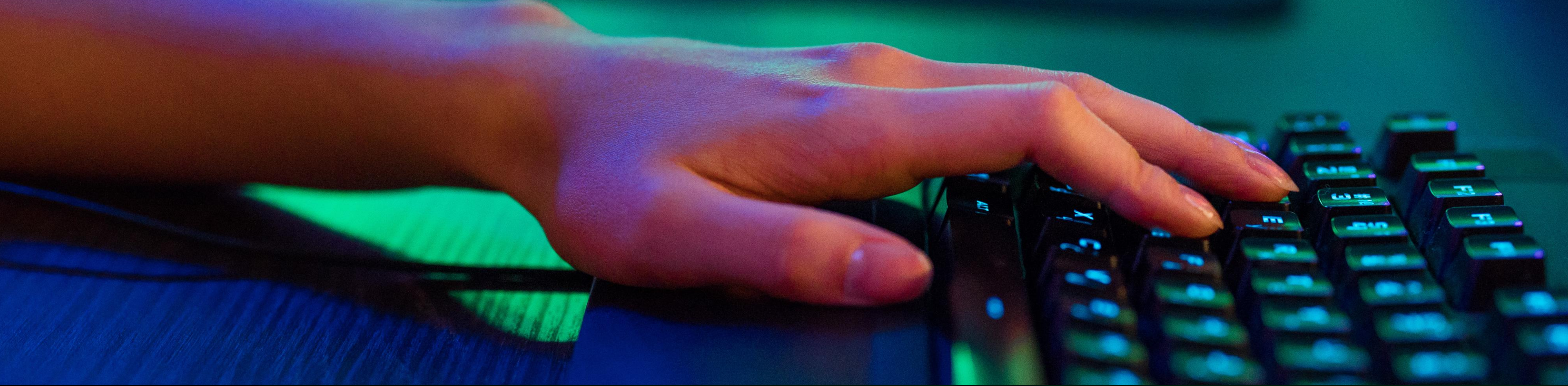
- Describe the scraping method used.
- Explain how the text was processed.
- Compare model performance with evaluation metrics.
- Discuss limitations and possible improvements.



SUBMISSION REQUIREMENTS

- . **Deadline:** One week.
- . **Format:** PDF or Word document.
- . **Grading Criteria:**
 - Data Collection (30%).
 - Model Training & Evaluation (40%).
 - Report Clarity & Analysis (30%).





THANK YOU

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